$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 1 \\ 1 & -1-\lambda \end{vmatrix} = 0$$

$$-(\lambda-1)^2 - 1 = 0$$

$$-(1-\lambda^2) - 1 = 0$$

$$\lambda = \pm\sqrt{2}$$

→ Eigenvalues of A^{13} is $\lambda^{13} = \pm 64\sqrt{2}$